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AY 2024-2025



**Computer Networks
Lab Record**

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Experiment 1: Study of different types of Network cables

Aim:

To study different types of Network cables and Practically implement the cross-wired cable and straight through cable using clamping tool.

Apparatus:

Twisted pair Cables.

Procedure:

Following should be understood for this practical.

i. Twisted pair Cable

A twisted pair consists of two insulated conductor twisted together in the shape of a spiral. It can be shielded or unshielded. The unshielded twisted pair cables are very cheap and easy to install. But they are very badly affected by the electromagnetic noise interference. Twisting of wires will reduce the effect of noise or external interference. The induced emf into the two wires due to interference tends to cancel each other due to twisting. Number of twists per unit length will determine the quality of cable. More twists means better quality.

There are 3 types of UTP cables:-

- 1) Straight-through cable
- 2) Crossover cable
- 3) Roll-over cable

A. Straight-through cable

Straight-Through refers to cables that have the pin assignments on each end of the cable. In other words Pin 1 connector A goes to Pin 1 on connector B, Pin 2 to Pin 2 etc. Straight-Through wired cables are most commonly used to connect a host to client. When we talk about cat5e patch cables, the Straight-Through wired cat5e patch cable is used to connect computers, printers and other network client devices to the router switch or hub (the host device in this instance).

B. Crossover cable

Crossover wired cables (commonly called crossover cables) are very much like Straight-Through cables with the exception that TX and RX lines are crossed (they are at opposite positions on either end of the cable. Using the 568-B standard as an example below you will see that Pin 1 on connector A goes to Pin 3 on connector B. Pin 2 on connector A goes to Pin 6 on connector B etc. Crossover cables are most commonly used to connect two hosts directly. Examples would be connecting a computer directly to another computer, connecting a switch directly to another switch, or connecting a router to a router. Note: While in the past when connecting two host devices directly a crossover cable was required. Now days most devices have auto sensing technology that detects the cable and device and crosses pairs when needed.

C. Roll-over cable

Rollover wired cables most commonly called rollover cables, have opposite Pin assignments on each end of the cable or in other words it is "rolled over". Pin 1 of connector A would be connected to Pin 8 of connector B. Pin 2 of connector A would be connected to Pin 7 of connector B and so on. Rollover cables, sometimes referred to as Host cables are most commonly used to connect to a devices console port to make programming changes to the device. Unlike crossover and straight-wired cables, rollover cables are not intended to carry data but instead create an interface with the device.

Diagrams:

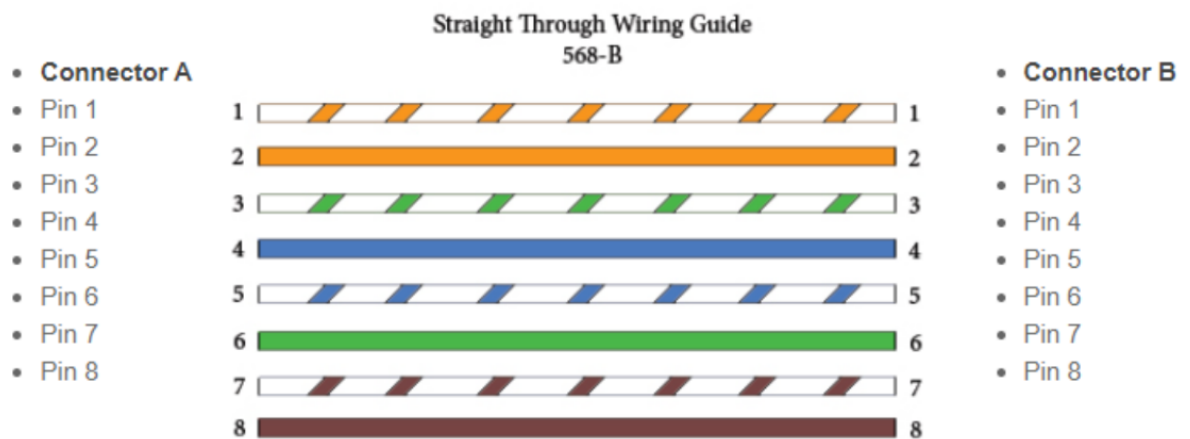


Figure 1: Diagram of Connection of pins in a Straight Through Cable

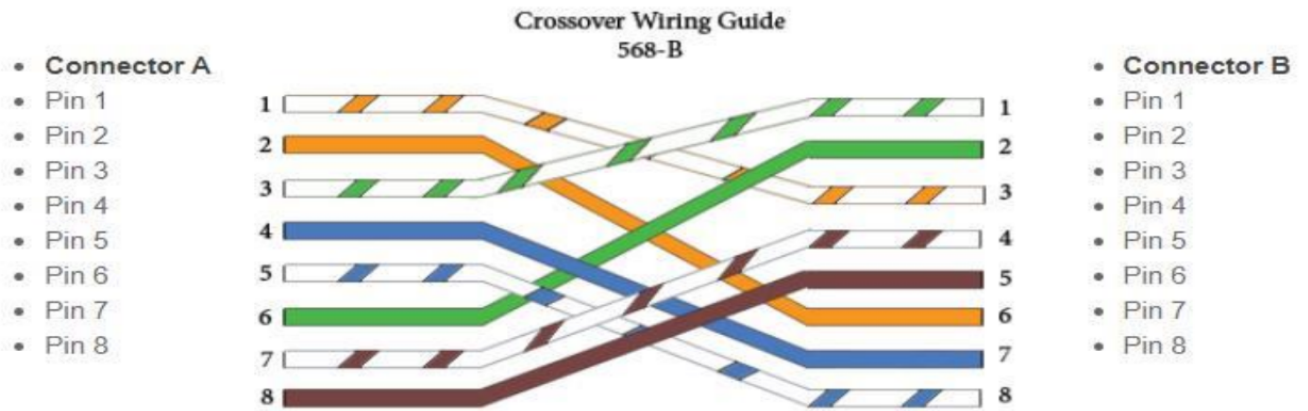


Figure 2: Diagram of Connection of pins in a Crossover Cable

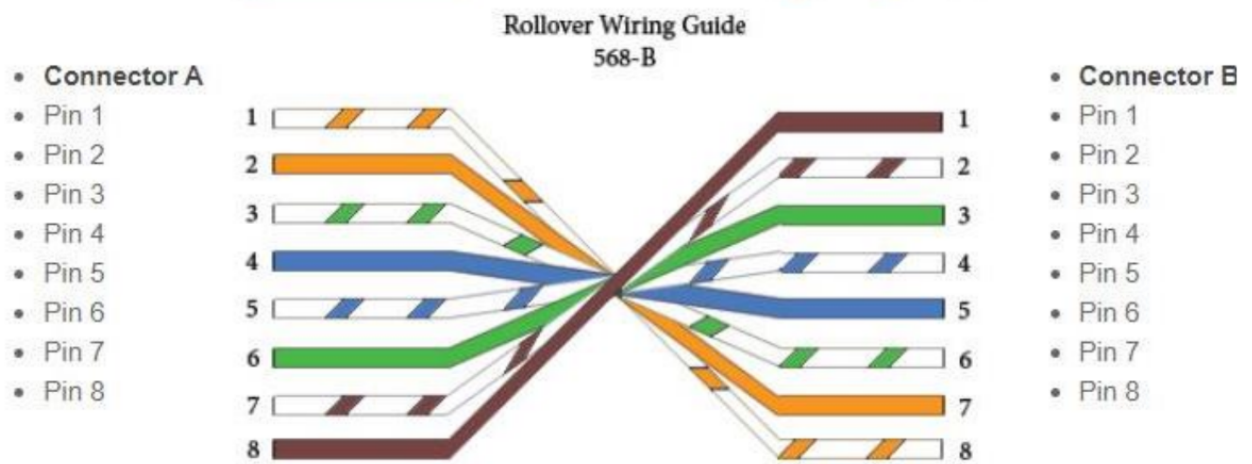


Figure 3: Diagram of Connection of pins in a Roll-over Cable

Experiment 2: Study of Network Devices in Detail

Aim:

To study the following Network Devices in Detail

- Repeater
- Hub
- Switch
- Bridge
- Router
- Gate Way

Apparatus:

No software or hardware needed.

Procedure:

Following should be understood for this practical.

1. Repeater:

Functioning at Physical Layer. A repeater is an electronic device that receives a signal and retransmits it at a higher level and/or higher power, or onto the other side of an obstruction, so that the signal can cover longer distances. Repeater have two ports, so cannot be use to connect for more than two devices.

2. Hub:

An Ethernet hub, active hub, network hub, repeater hub, hub or concentrator is a device for connecting multiple twisted pair or fiber optic Ethernet devices together and making them act as a single network segment. Hubs work at the physical layer (layer 1) of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

3. Switch:

A network switch or switching hub is a computer networking device that connects network segments. The term commonly refers to a network bridge that processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are often referred to as Layer 3 switches or multilayer switches.

4. Bridge:

A network bridge connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet networks, the term bridge formally means a device that behaves according to the IEEE 802.1D standard. A bridge and switch are very much alike; a switch being a bridge with numerous ports. Switch or Layer 2 switch is often used interchangeably with bridge. Bridges can analyze incoming data packets to determine if the bridge is able to send the given packet to another segment of the network.

5. Router:

A router is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. Each data packet contains address information that a router can use to determine if the source and destination are on the same network, or if the data packet must be transferred from one network to another. Where multiple routers are used in a large collection of interconnected networks, the routers exchange information about target system addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected networks.

6. Gate Way:

In a communications network, a network node equipped for interfacing with another network that uses different protocols.

- A gateway may contain devices such as protocol translators, impedance matching devices, rate converters, fault isolators, or signal translators as necessary to provide system interoperability. It also requires the establishment of mutually acceptable administrative procedures between both networks.
- A protocol translation/mapping gateway interconnects networks with different network protocol technologies by performing the required protocol conversions.

Conclusion:

The study of network devices is fundamental to understanding how modern computer networks operate. These devices, such as routers, switches, hubs, and firewalls, play critical roles in ensuring efficient data transmission, security, and management across a network. By examining each device's function, we appreciated how they interconnect to form complex, scalable, and secure networks.

Experiment 3: Study of network IP

Aim:

To study network IP

- Classification of IP address
- Subnetting
- Supernetting

Apparatus:

No software or hardware needed.

Procedure:

Following should be understood for this practical.

Classification of IP address

Class	Address Range	Supports
Class A	1.0.0.1 to 126.255.255.254	Supports 16 million hosts on each of 127 networks.
Class B	128.1.0.1 to 191.255.255.254	Supports 65,000 hosts on each of 16,000 networks.
Class C	192.0.1.1 to 223.255.254.254	Supports 254 hosts on each of 2 million networks.
Class D	224.0.0.0 to 239.255.255.255	Reserved for multicast groups.
Class E	240.0.0.0 to 254.255.255.254	Reserved.

Subnetting

- **Efficient IP Address Allocation:** Subnetting reduces wastage of IP addresses by dividing a large network into smaller, manageable subnetworks.
- **Improved Network Management:** It simplifies network management by segregating traffic for different departments or geographic locations.
- **Enhanced Security:** It isolates network segments, reducing the risk of unauthorized access to sensitive data.
- **Reduces Network Traffic:** Limits broadcast traffic within a subnet, improving overall network performance.

How to Calculate Subnet Mask:

1. Start with the default mask:
 - Class A: 255.0.0.0
 - Class B: 255.255.0.0
 - Class C: 255.255.255.0
2. Borrow bits from the host portion to create additional subnets.
3. Number of subnets = 2^n , where n is the number of borrowed bits.
4. Hosts per subnet = $2^h - 2$, where h is the number of host bits.

How to Identify Subnet Address:

- Convert the IP address to binary.
- Apply the subnet mask using a bitwise AND operation.
- The resulting binary value is the subnet address.

Example:

- IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.240 (28 bits, creating 16 subnets)
- Subnet Address: 192.168.1.16

Supernetting

Why We Develop Supernetting:

- Reduces the size of routing tables by aggregating multiple networks into a single larger block.
- Optimizes routing decisions and enhances performance in backbone networks.
- Efficiently utilizes IP space by combining smaller subnets into a larger block.

How to Calculate Supernet Mask:

1. Identify the range of IP addresses to be aggregated.
2. Determine the common bits in their binary representation.
3. The number of common bits becomes the prefix length for the supernet mask.

Example:

- Networks: 192.168.0.0/24 and 192.168.1.0/24

- Binary Representation:
 - 192.168.0.0: 11000000.10101000.00000000.00000000
 - 192.168.1.0: 11000000.10101000.00000001.00000000
- Common Bits: 23
- Supernet Mask: 255.255.254.0 (/23)

How to Identify Supernet Address:

- Identify the smallest and largest addresses in the range.
- Convert these to binary.
- Find the common bits and set all host bits to 0.

Example:

- Aggregated Networks: 192.168.0.0/24 and 192.168.1.0/24
- Supernet Address: 192.168.0.0
- Supernet Mask: /23

Summary Table

Concept	Subnetting	Supernetting
Purpose	Divide a large network into smaller ones.	Combine multiple smaller networks into one.
Efficiency	Reduces broadcast traffic.	Reduces routing table size.
Mask Calculation	Borrow bits from the host portion.	Use common bits from contiguous networks.
Example Mask	255.255.255.240 (/28)	255.255.254.0 (/23)

Conclusion:

The study of network IP (Internet Protocol) is essential for understanding how devices communicate over a network and across the internet. IP addresses serve as unique identifiers for devices, enabling them to send and receive data accurately. We understood the foundation of Network IP and managing networks.

Experiment 4: Connect the computers in Local Area Network

Aim:

To connect the computers in Local Area Network

Apparatus:

Cisco Packet Tracer.

Procedure:

Network Topologies

1. Star Topology:

- Devices are connected to a central device such as a switch or hub.
- If the central device fails, the entire network goes down.
- Easy to add or remove devices without affecting the network.
- Suitable for small networks like offices or schools.

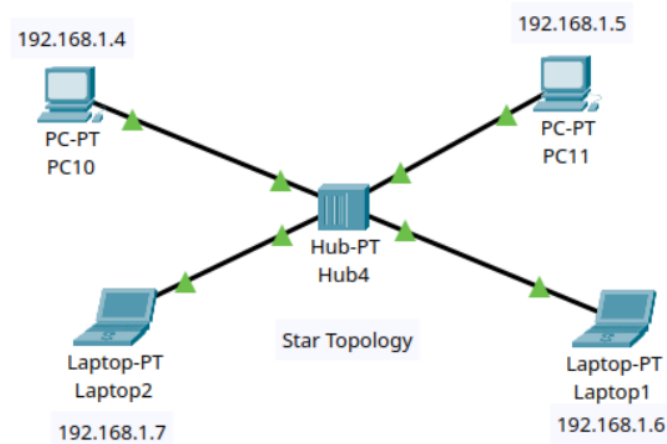


Figure 4: Diagram of Star Topology

2. Bus Topology:

- Devices share a single communication line, called a bus.
- Cost-effective for small networks but not scalable.

- If the backbone fails, the entire network is disrupted.
- Mostly used in legacy or experimental setups.

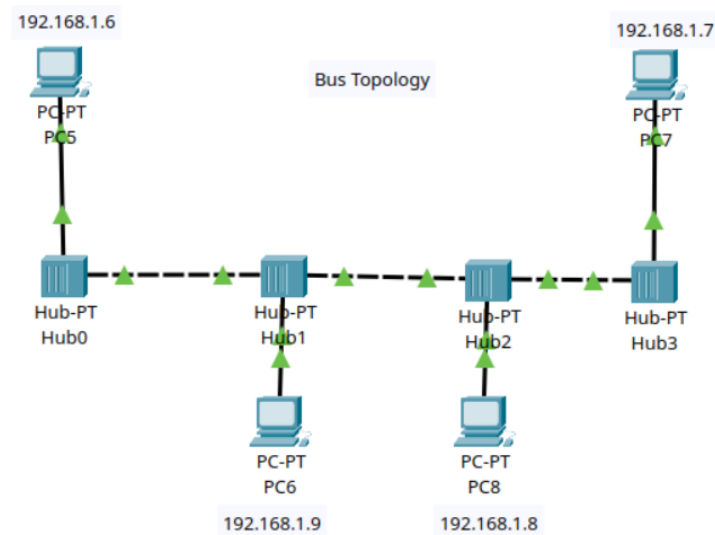


Figure 5: Diagram of Bus Topology

3. Ring Topology:

- Devices are connected in a closed loop where data travels in one direction.
- Easy to troubleshoot as the fault location is identifiable.
- Not commonly used due to the dependency on a single path for data transmission.
- Useful in token-based communication systems like FDDI.

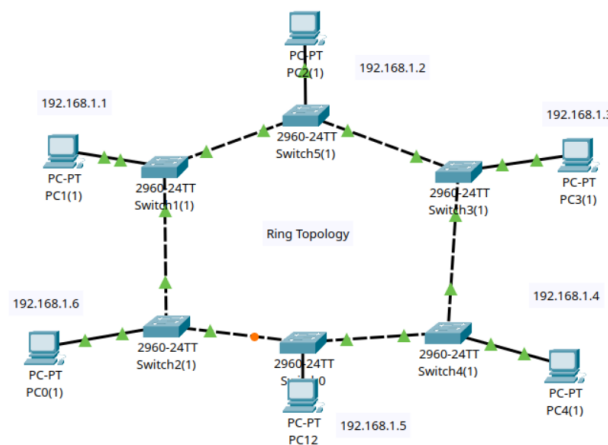


Figure 6: Diagram of Ring Topology

4. Mesh Topology:

- Every device is connected to every other device, providing multiple paths.
- High fault tolerance and redundancy, as no single point of failure exists.
- Expensive and complex to set up due to a large number of connections.
- Used in mission-critical systems like data centers or military networks.

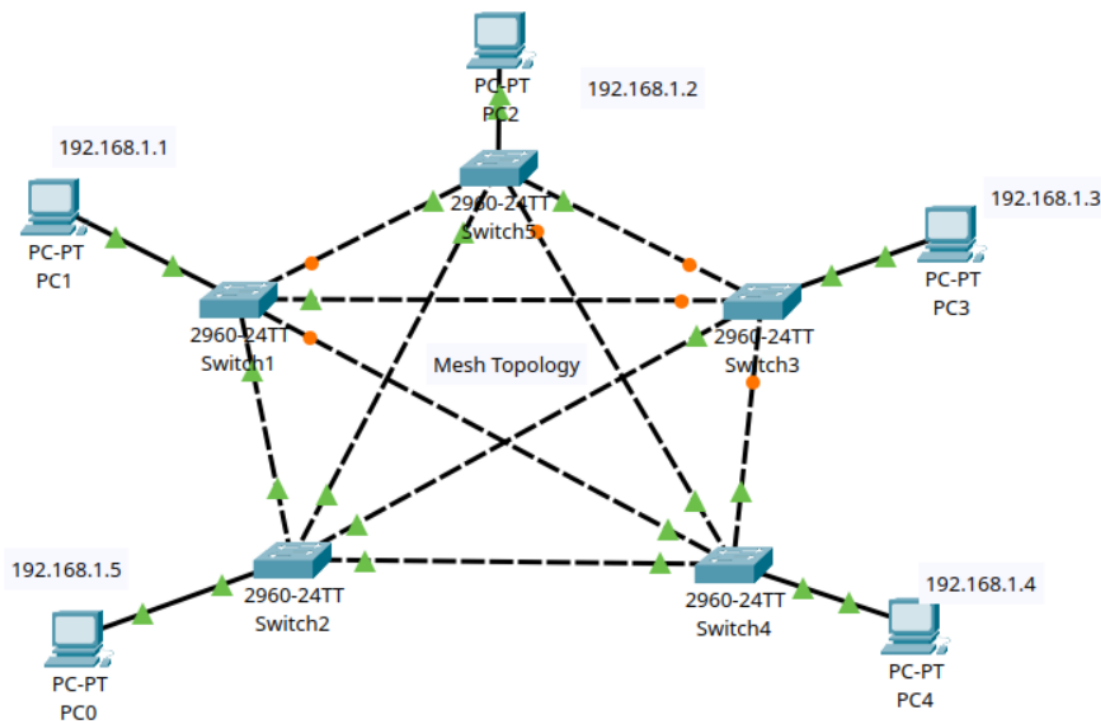


Figure 7: Diagram of Mesh Topology

5. Hybrid Topology:

- Combines two or more topologies (e.g., Star and Bus).
- Provides the benefits of multiple topologies and offers flexibility.
- Can be customized to meet specific network requirements.
- Common in enterprise networks with varied needs.

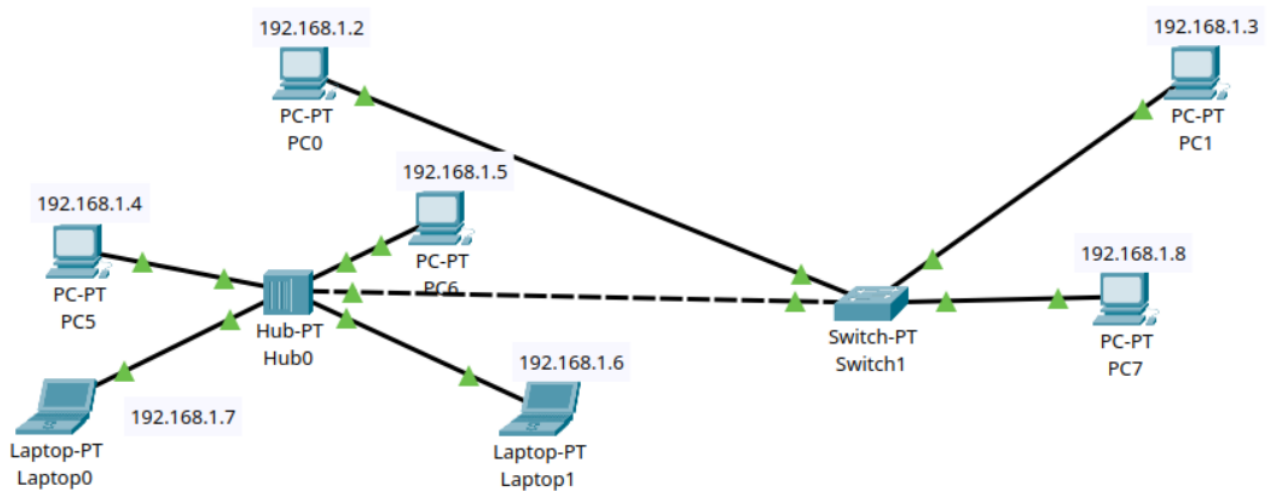


Figure 8: Diagram of Hybrid Topology

Conclusion:

Topology	Description	Example Use Case
Star	Devices connect to a central device (e.g., switch).	Offices, small networks.
Bus	Devices connect along a single backbone.	Small, legacy networks.
Ring	Devices connect in a closed loop.	Token-based systems.
Mesh	Every device connects to every other device.	High-redundancy networks.
Hybrid	Combines two or more topologies.	Enterprise networks with varied needs.

Experiment 5: Study of basic network command and Network configuration commands

Aim:

To study and understand basic network commands and network configuration commands.

Apparatus (Software):

Cisco Packet Tracer.

Procedure

To perform this experiment, follow these steps:

1. Understand basic networking commands such as `ping`, `tracert`, etc.
2. Execute and understand the following types of commands:
 - `ping`
 - `tracert`
 - `nslookup`
 - `pathping`

Details of Commands

ping Command:

`ping` sends an ICMP ECHO_REQUEST packet to a specified host. If the host responds, an ICMP packet is returned. This is used to check whether a machine is active or reachable on the network.

Traceroute:

`tracert` is a command which can show you the path a packet of information takes from your computer to one you specify. It will list all the routers it passes through until it reaches its destination, or fails to and is discarded. In addition to this, it will tell you how long each 'hop' from router to router takes.

nslookup:

`tracert` Displays information from Domain Name System (DNS) name servers. NOTE :If you write the command as above it shows as default your pc's server name firstly.

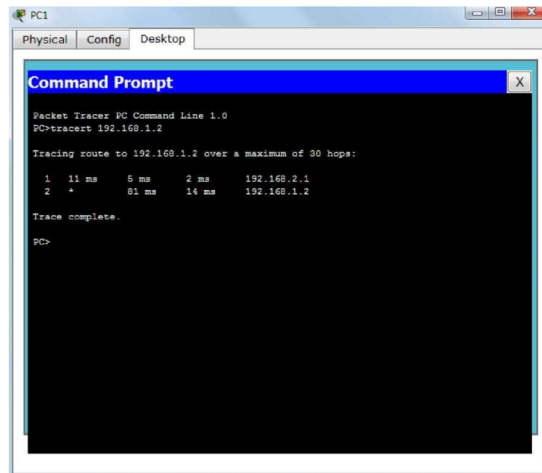


Figure 9: Command tracert

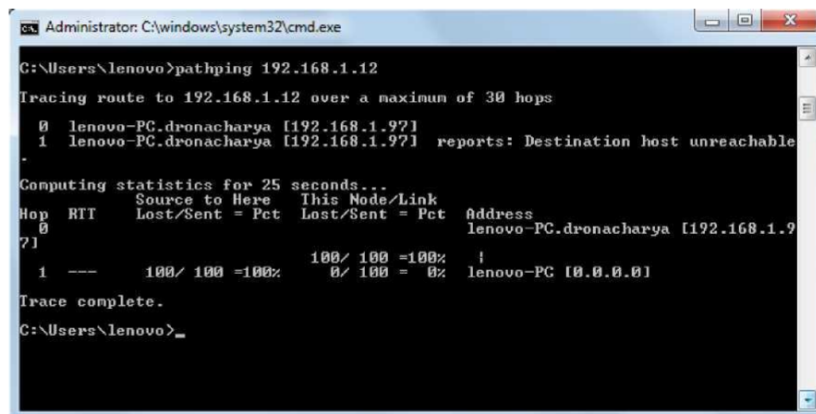


Figure 10: Command pathping

pathping:

pathping A better version of **tracert** that gives you statistics about packet loss and latency.

Conclusion

This experiment familiarizes essential network commands. Understanding these commands is vital for managing and maintaining modern networks.

Experiment 6: Performing an Initial Switch Configuration

Aim:

To perform an Initial Switch Configuration

Apparatus:

Cisco Packet Tracer:- Catalyst 2960 Switch, PC, RS 232 Cable.

Objectives

- Perform an initial configuration of a Cisco Catalyst 2960 switch.

Background / Preparation

In this experiment, we will configure these settings on the our Cisco Catalyst 2960 switch:

- Host name
- Console password
- vty password
- Privileged EXEC mode password
- Privileged EXEC mode secret

Step 1: Configure the Switch Host Name

- a. From the PC, use a console cable and connect to the console of Cisco Catalyst 2960 switch.
- b. Set the host name on the switch to **Abdul Gafoor** using these commands:

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Abdul_Gafoor
```

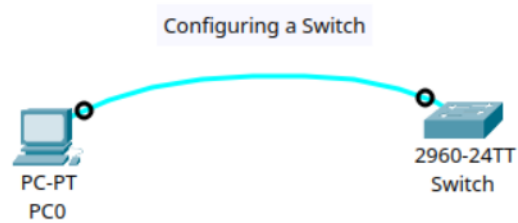


Figure 11: Connecting Console cable to Switch

Step 2: Configure the Privileged Mode Password and Secret

- a. From global configuration mode, configure the password as `cisco`.

```
Abdul_Gafoor(config)#enable password cisco
```

- b From global configuration mode, configure the secret as `cisco123`.

```
Abdul_Gafoor(config)#enable secret cisco123
```

Step 3: Configure the Console Password

- a. From global configuration mode, switch to configuration mode to configure the console line.

```
Abdul_Gafoor(config)#line console 0
```

- b From line configuration mode, set the password to `cisco` and require the password to be entered at login.

```
Abdul_Gafoor(config-line)#password cisco
Abdul_Gafoor(config-line)#login
Abdul_Gafoor(config-line)#exit
```

Step 4: Verify the Configuration

To verify the configuration, use the following commands:

- a. Display the configured hostname to ensure it is set to `Abdul Gafoor`.

```
Abdul_Gafoor#show running-config
```

- b Verify that the privileged mode password and secret are configured correctly.

```
Abdul_Gafoor#show running-config | include enable
```

- c Check the console password by attempting to log out and back into the console.

```
Abdul_Gafoor>logout
```

Press RETURN to get started.

Password: (Enter configured password: cisco)

Conclusion:

The Switch Catalyst 2960 (Hostname: Abdul Gafoor) is Configured.

Experiment 7: Performing an Initial Router Configuration

Aim:

To perform an Router Switch Configuration

Apparatus:

Cisco Packet Tracer:- Pc, Router, RS 232 Cable.

Objectives

- Perform an initial configuration of a Cisco Router.

Background / Preparation

In this experiment, we will configure these settings on a Cisco Router:

- Host name
- Console password
- vty password
- Privileged EXEC mode password
- Privileged EXEC mode secret

Step 1: Configure the Router Host Name

- a. From the Customer PC, use a console cable and connect to Cisco Router.
- b. Set the host name on the switch to **Abdul Gafoor** using these commands:

```
Router>enable
Router#configure terminal
Router(config)#hostname Abdul_Gafoor
```

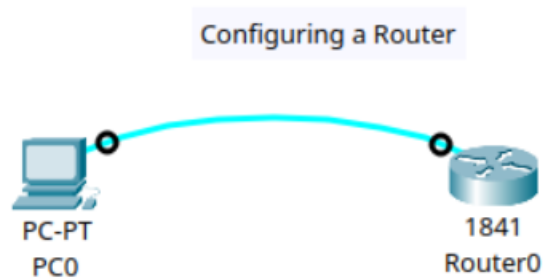


Figure 12: Connecting Console cable to Router

Step 2: Configure the Privileged Mode Password and Secret

- a. From global configuration mode, configure the password as `cisco`.

```
Abdul_Gafoor(config)#enable password cisco
```

- b. From global configuration mode, configure the secret as `cisco123`.

```
Abdul_Gafoor(config)#enable secret cisco123
```

Step 3: Configure the Console Password

- a. From global configuration mode, switch to configuration mode to configure the console line.

```
Abdul_Gafoor(config)#line console 0
```

- b. From line configuration mode, set the password to `cisco` and require the password to be entered at login.

```
Abdul_Gafoor(config-line)#password cisco
Abdul_Gafoor(config-line)#login
Abdul_Gafoor(config-line)#exit
```

Step 4: Verify the Configuration

To verify the configuration, use the following commands:

- a. Display the configured hostname to ensure it is set to **Abdul Gafoor**.

```
Abdul_Gafoor#show running-config
```

- b. Verify that the privileged mode password and secret are configured correctly.

```
Abdul_Gafoor#show running-config | include enable
```

- c. Check the console password by attempting to log out and back into the console.

```
Abdul_Gafoor>logout
```

Press RETURN to get started.

Password: (Enter configured password: cisco)

Conclusion:

The Cisco Router (Hostname: Abdul Gafoor) is Configured.

Experiment 8: Configuring and Troubleshooting a Switched Network

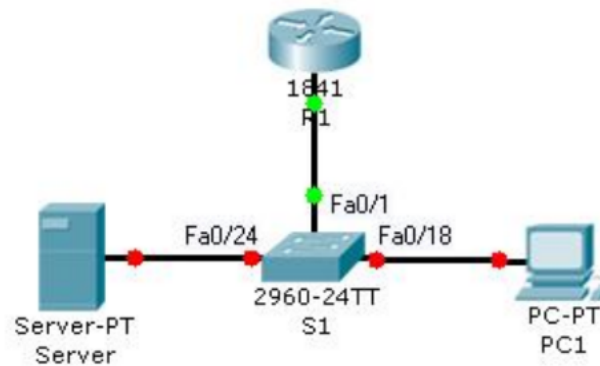


Figure 13: Configuring and Troubleshooting a Switched Network

Objectives:

- Establish console connection to the switch.
- Configure the host name and VLAN1.
- Use the help feature to configure the clock.
- Configure passwords and console/Telnet access.
- Configure login banners.
- Configure the router.
- Solve duplex and speed mismatch problems.
- Configure port security.
- Secure unused ports.
- Manage the switch configuration file.

Background / Preparation:

In this Packet Tracer Skills Integration Challenge activity, you will configure basic switch management, including general maintenance commands, passwords, and port security. This activity provides you an opportunity to review previously acquired skills.

Step 1: Establish Console Connection to the Switch

1. Connect a console cable from your PC to the console port on the switch.

Step 2: Configure the Host Name and VLAN1

1. Enter privileged EXEC mode using the `enable` command.
2. Enter global configuration mode using `configure terminal`.
3. Set the host name:

- `hostname Abdul Gafoor`

4. Configure the VLAN1 interface:

- `interface vlan 1`
- `ip address 192.168.1.2 255.255.255.0`
- `no shutdown`

Step 3: Use the Help Feature to Configure the Clock

1. In global configuration mode, type `clock ?` to view clock configuration options.
2. Set the current date and time using the `clock set` command.

Step 4: Configure Passwords and Console/Telnet Access

1. Set the console password:
2. Set the VTY (Telnet) password:

- `line console 0`
- `password cisco`
- `login`
- `line vty 0 4`
- `password cisco`
- `login`

Step 5: Configure Login Banners

1. Configure a banner message of the day:
- `banner motd # Unauthorized access is prohibited! #`

Step 6: Configure the Router

1. Access the router and configure basic settings, including IP addressing and interface activation.
2. Verify the configuration with `show ip interface brief`.

Step 7: Solve Duplex and Speed Mismatch Problems

1. Check interface settings using `show interfaces`.
2. Manually configure duplex and speed settings if mismatches are detected:
 - `duplex full`
 - `speed 100`

Step 8: Configure Port Security

1. Configure port security on an interface:
 - `interface FastEthernet0/1`
 - `switchport mode access`
 - `switchport port-security`
 - `switchport port-security maximum 2`
 - `switchport port-security violation shutdown`
 - `switchport port-security mac-address sticky`

Step 9: Secure Unused Ports

1. Disable unused ports to enhance security:
 - `interface range FastEthernet0/2-24`
 - `shutdown`

Step 10: Manage the Switch Configuration File

1. Save the configuration to NVRAM:
 - `copy running-config startup-config`
2. Verify the saved configuration:
 - `show startup-config`

Conclusion:

Configured and Troubleshooted a Switched Network

Experiment 9: Configuring WEP on a Wireless Router

Aim:

To Configure WEP (Wired Equivalent Privacy) on a Wireless Router.

Apparatus:

Cisco Packet Tracer.

Objectives:

- Configure WEP security between a workstation and a wireless router.

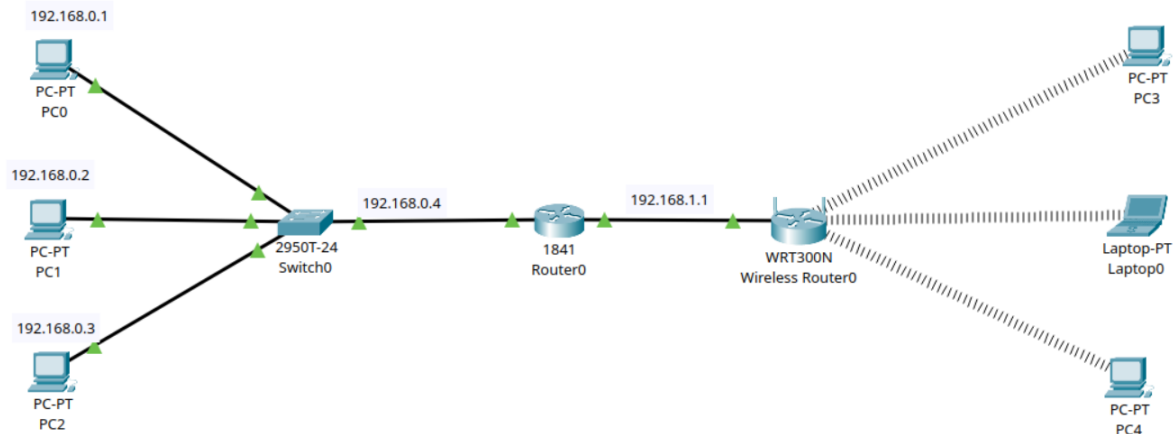


Figure 14: Wired Equivalent Privacy [WEP]

Step 1: Configure the Wireless Router to Require WEP

- Click the Wireless Router icon. Then, click the GUI tab to access the router's web management interface.
- Click the **Wireless** menu option and change the Network Name (SSID) from **Default** to **CSE-C1**. Leave the other settings with their default options.
- Click the **Save Settings** button at the bottom of the **Basic Wireless Settings** window.
- Click the **Wireless Security** submenu under the **Wireless** menu to display the current wireless security parameters.

- e. From the **Security Mode** drop-down menu, select **WEP**.
- f. In the **Key1** text box, type **12345FFFFFF**. This will be the new WEP pre-shared key to access the wireless network.
- g. Click the **Save Settings** button at the bottom of the **Wireless Security** window.

Step 2: Configure WEP on the PC

- a. Click on the **PC**.
- b. Click the **Config** tab.
- c. Click the **Wireless** button to display the current wireless configuration settings on the workstation.
- d. Change the **SSID** to **CSE-C1**.
- e. Change the **Security Mode** to **WEP**. Enter **12345FFFFFF** in the **Key** text box, and then close the window.

Step 3: Verify the Configuration

After you configure the correct WEP key and SSID on the customer wireless workstation, notice that there is a wireless connection between the workstation and the wireless router.

- a. Click on the **PC**.
- b. Click the **Desktop** tab to view the applications that are available.
- c. Click on the **Command Prompt** application to bring up the command prompt.
- d. Type **ipconfig /all** and press **Enter** to view the current network configuration settings.
- e. Type **ping 192.168.2.1** to verify connectivity to the LAN interface of the customer wireless router.
- f. Close the command prompt window.
- g. Open a web browser.
- h. In the address bar of the web browser window, type **http://192.168.1.10**. Press **Enter**. The Intranet web page that is running on the customer server appears. You have just verified that the customer wireless workstation has connectivity to the rest of the customer network.
- i. Click the **Check Results** button at the bottom of this instruction window to check your work.

Experiment 10: Using the Cisco IOS Show Commands

Objective:

- Use the Cisco IOS show commands.

Background / Preparation:

The Cisco IOS show commands are used extensively when working with Cisco equipment. In this activity, you will use the show commands on a router that is located at an ISP.

Step 1: Connect to the ISP Cisco 1841 Router

- Use the terminal emulation software on the ISP PC to connect to the Cisco 1841 router.
- The `ISPRouter>` prompt indicates that you are in user EXEC mode.
- Type `enable` at the prompt. The `ISPRouter#` prompt indicates that you are in privileged EXEC mode.

Step 2: Explore the Show Commands

Use the information displayed by these show commands to answer the questions in the Reflection section.

- a. Type `show arp`. This command displays the Address Resolution Protocol (ARP) table for the router.
- b. Type `show flash`. This command shows the contents of the router's flash memory.
- c. Type `show ip route`. This command displays the IP routing table.
- d. Type `show interfaces`. This command provides detailed information about the status of each interface on the router.
- e. Type `show protocols`. This command shows the status of all configured network protocols, such as IP and routing protocols.
- f. Type `show users`. This command displays a list of users currently logged into the router.
- g. Type `show version`. This command displays the router's software version, hardware configuration, and other details.

Experiment 11: Examining WAN Connections

Objective:

The show commands are very powerful commands for troubleshooting and monitoring networks. They give a static image of the network at a given time. The use of a variety of show commands will give a clear picture of how the networking is communicating and transferring data.

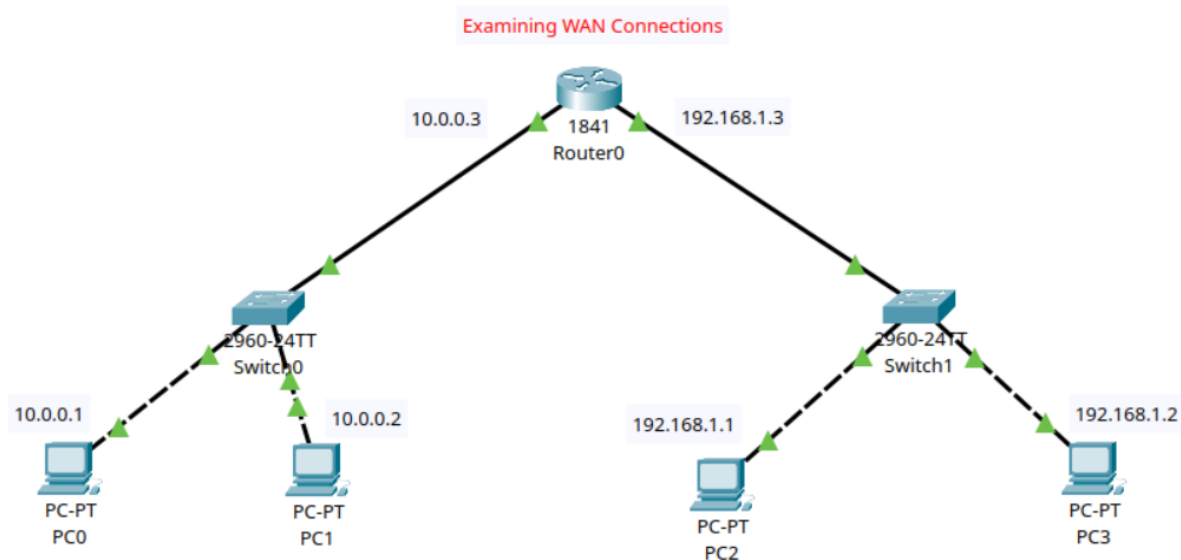


Figure 15: Examining WAN Connections

Background / Preparation:

The physical topology of the network has been designed using Frame Relay. To test the network connectivity, use a variety of show commands.

Step 1: Examine the Configuration of Branch1 and Branch2

- Click on Branch1 and use various show commands to view the connectivity to the network.
- Use the `show running-configuration` command to view the router configuration.
- Use the `show ip interface brief` command to view the status of the interfaces.
- Click on Branch2 and use various show commands to view the connectivity to the network.

- e. Use the `show running-configuration` command to view the router configuration.
- f. Use the `show ip interface brief` command to view the status of the interfaces.

Step 2: Examine the Configuration of Main

- a. Click on Main and use a variety of show commands to view the connectivity to the network.
- b. Use the `show running-configuration` command to view the router configuration.
- c. Use the `show ip interface brief` command to view the status of the interfaces.

Conclusion:

Examined WAN Connections.

Experiment 12: Interpreting Ping and Traceroute Output

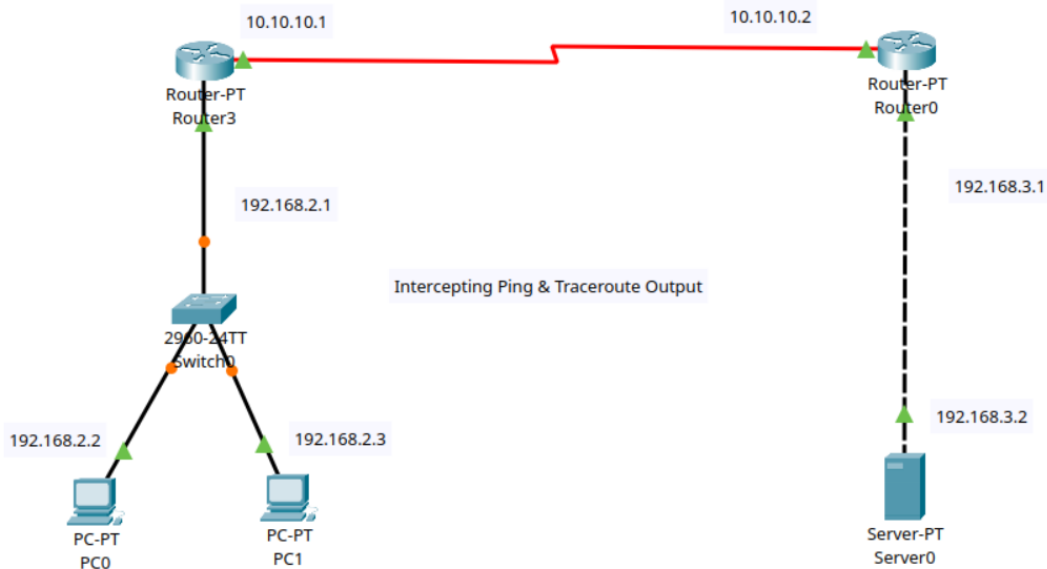


Figure 16: Interpreting Ping and Traceroute Output

Objectives:

- Distinguish the difference between successful and unsuccessful ping attempts.
- Distinguish the difference between successful and unsuccessful traceroute attempts.

Background / Preparation:

In this experiment, you will test end-to-end connectivity using ping and traceroute. At the end of this experiment, you will be able to distinguish the difference between successful and unsuccessful ping and traceroute attempts.

Step 1: Test Connectivity Using Ping from a Host Computer and a Router

- Click N-Host, click the Desktop tab, and then click Command Prompt.
- From the Command Prompt window, ping the Cisco server at www.cisco.com.

Packet Tracer PC Command Line 1.0

```
PC>ping www.cisco.com
Pinging 64.100.1.185 with 32 bytes of data:
```

```
Request timed out.  
Reply from 64.100.1.185: bytes=32 time=185ms TTL=123  
Reply from 64.100.1.185: bytes=32 time=281ms TTL=123  
Reply from 64.100.1.185: bytes=32 time=287ms TTL=123  
Ping statistics for 64.100.1.185:  
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 185ms, Maximum = 287ms, Average = 251ms  
PC>
```

From the output, you can see that N-Host was able to obtain an IP address for the Cisco server. The IP address was obtained using DNS. Also, notice that the first ping failed. This failure is most likely due to the lack of ARP convergence between the source and destination. If you repeat the ping, you will notice that all pings succeed.

- From the Command Prompt window on N-Host, ping E-Host at 192.168.4.10.
- The pings fail. If you do not want to wait for all four unsuccessful ping attempts, press **Ctrl+C** to abort the command, as shown below.

Packet Tracer PC Command Line 1.0

```
PC>ping 192.168.3.2  
Pinging 192.168.3.2 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Ping statistics for 192.168.3.2:  
Packets: Sent = 3, Received = 0, Lost = 3 (100% loss)
```

Conclusion:

Interpreted Ping and Traceroute Outputs.

Experiment 13: Demonstrating Distribution Layer Functions

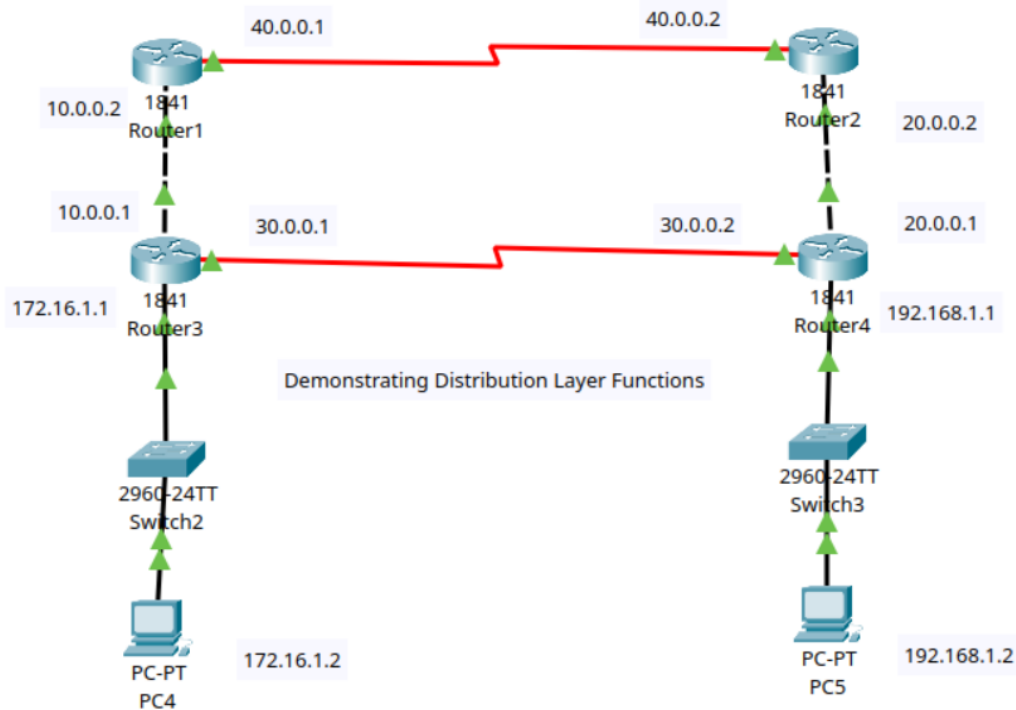


Figure 17: Demonstrating Distribution Layer Functions

Objective:

- Demonstrate the functions performed by the Distribution Layer devices.

Background / Preparation

VLANs can be added to a network for security purposes and traffic control. Devices on separate VLANs are unable to communicate unless a router has been configured to help with this communication. Observe how packet filtering and route summarization traverse the network using simulation mode.

Step 1: Setup Simulation Filters to Capture Routing Protocols

- Enter simulation mode in Packet Tracer.
- Click on the **Edit Filters** button.
- Select EIGRP.

- d. Click on the **Reset Simulation** button.
- e. Click **Auto Capture/Play**.
- f. Observe the EIGRP updates.

Step 2: Test Connectivity Between the Network Devices Using Realtime Mode

- a. From PC0 ping PC1, PC2, PC3, and PC4.
- b. From PC1 ping PC0, PC2, PC4, PC3.

Step 3: Test Connectivity Between the Network Devices Using Simulation Mode

- a. Switch from Realtime mode to Simulation mode.
- b. Create a simple PDU from PC0 to PC1. Click **Capture/Forward** until the PDU has made the complete trip to PC1 and back.
- c. In the **Event List** view the PDU events.
- d. Create another PDU from PC0 to PC2.

Conclusion:

Demonstrating Distribution Layer Functions.

Experiment 14: Placing ACLs

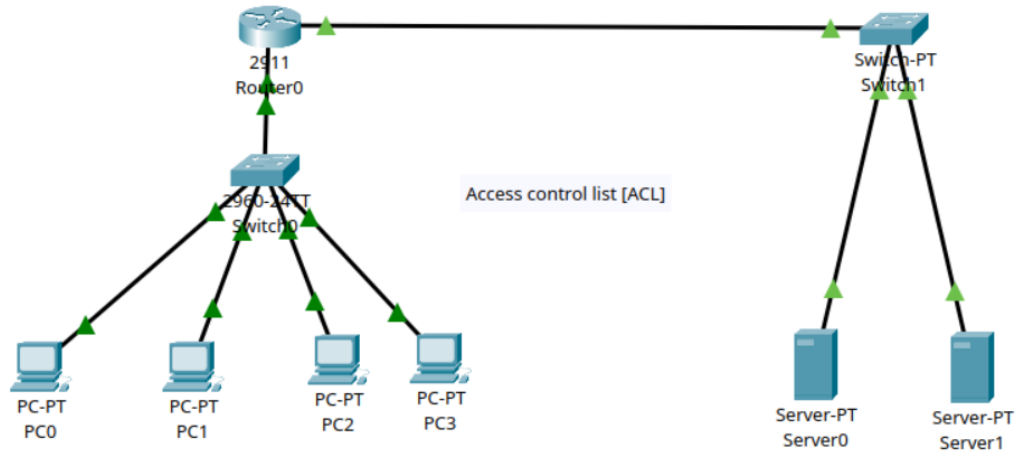


Figure 18: Access Control List

Objectives:

- Verify network connectivity.
- Examine the Access Control Lists (ACLs) that are configured on the routers.
- Determine the appropriate interface to apply the ACLs.
- Examine the effects of the ACL.

Background / Preparation:

This activity demonstrates how the flow of network traffic is affected by applying an ACL to permit or deny traffic in the network. The network administrator has decided that all external web traffic goes only to the Web server. Also, in order to protect the data of their employees. Therefore, ACLs will need to be implemented on the network.

Step 1: Verify Network Connectivity

- a. Verify that all of the PCs can communicate with each other and with the servers.
- b. Verify that the Internet Host can access the Web server (192.168.0.3), Sales server (192.168.10.2), and HR server (192.168.40.2) using the browser.

Step 2: Examine the Access Control Lists that are Configured on the Routers

- a. Access the **Distribution1** router. Use the following commands to view the ACL that has been configured on the **Distribution1** router:
 - `show running-config`
 - `show access-lists 1`
- b. Access the **Gateway** router. Use the following commands to view the ACL that has been configured on the **Gateway** router:
 - `show running-config`
 - `show access-lists 100`

Step 3: Determine the Appropriate Interface to Apply the ACLs

- a. After examining the ACLs, determine on which interface the ACLs should be applied.
- b. The ACL must be applied to an interface or subinterface before it will affect the network traffic.
- c. The extended ACL should be placed closest to the source and the standard ACL should be closest to the destination.
- d. Remember that only one ACL per port, per protocol, per direction is allowed.
- e. Apply the ACL to the appropriate interface or subinterface.

Step 4: Examine the Effects of the ACL

- a. The Internet Host should be able to ping any device in the network, except HR1 or HR server.
- b. The Internet Host should be able to access the Web server (192.168.0.3) using the browser.
- c. The Internet Host should not be able to access either the HR server (192.168.40.1) or Sales server (192.168.10.2) using the browser.
- d. HR2 should be able to access the HR server (192.168.40.1) using ping or the browser.
- e. RandD2 should not be able to access the HR server (192.168.40.1) using ping or the browser.

Conclusion:

Successfully Placed ACLs.

Experiment 15: Examining Network Address Translation (NAT)

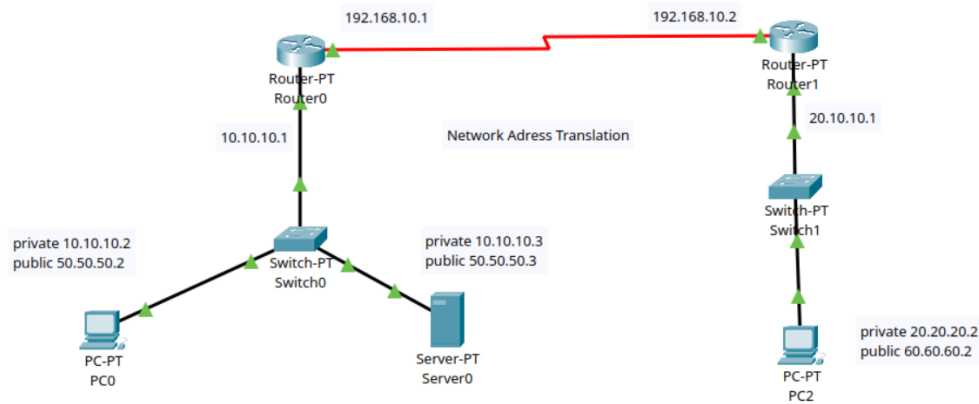


Figure 19: Network Address Translation

Objectives:

- Understand the purpose of NAT in a network.
- Configure NAT on a router to allow internal hosts to access the internet.
- Verify the NAT configuration.

Background / Preparation:

Network Address Translation (NAT) is a method used to map private IP addresses to a public IP address (or range of public IP addresses) to enable internet connectivity. In this activity, you will configure NAT on a router and verify its operation.

Step 1: Configure Basic Router Settings

1. Access the router's command-line interface (CLI).
2. Enter global configuration mode using the command:

```
configure terminal
```

3. Configure the interfaces:

- Assign IP addresses to the interfaces connecting the internal network and the ISP.

- Example:

```
interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface Serial0/0/0
ip address 203.0.113.1 255.255.255.252
no shutdown
```

Step 2: Define the NAT Inside and Outside Interfaces

1. Designate the inside and outside NAT interfaces:

```
interface FastEthernet0/0
ip nat inside

interface Serial0/0/0
ip nat outside
```

Step 3: Configure a NAT Pool

1. Define a NAT pool for public IP addresses:

```
ip nat pool PUBLIC_POOL 203.0.113.2 203.0.113.10 netmask 255.255.255.252
```

2. Create an access control list (ACL) to identify traffic for NAT:

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

Step 4: Configure Dynamic NAT

1. Link the NAT pool to the ACL:

```
ip nat inside source list 1 pool PUBLIC_POOL
```

Step 5: Verify the NAT Configuration

1. Use the following commands to verify NAT:

```
show ip nat translations
show ip nat statistics
```

2. Test connectivity by pinging an external address from a device in the internal network.

Experiment 16: Observing Static and Dynamic Routing

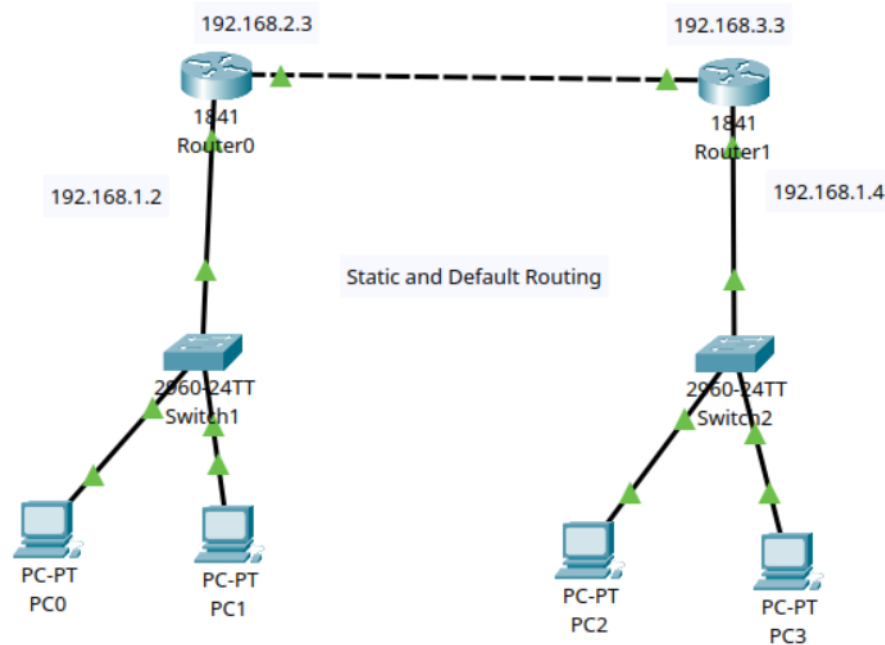


Figure 20: Static and Dynamic Routing

Objectives:

- Understand and configure static routing.
- Understand and configure dynamic routing using RIP.
- Verify the configuration for both routing methods.

Background / Preparation

Routing enables devices on different networks to communicate with one another. Static routing requires manual configuration, while dynamic routing protocols like RIP enable automatic route discovery and maintenance.

Static Routing Configuration

Step 1: Configure Basic Router Settings

1. Access the router's CLI and enter global configuration mode:

```
configure terminal
```

2. Assign IP addresses to the interfaces:

```
interface FastEthernet0/0  
ip address 192.168.1.1 255.255.255.0  
no shutdown
```

```
interface Serial0/0/0  
ip address 10.1.1.1 255.255.255.252  
no shutdown
```

Step 2: Configure Static Routes

1. Add static routes to reach remote networks:

```
ip route 192.168.2.0 255.255.255.0 10.1.1.2  
ip route 192.168.3.0 255.255.255.0 10.1.1.2
```

2. Verify the configuration using:

```
show ip route
```

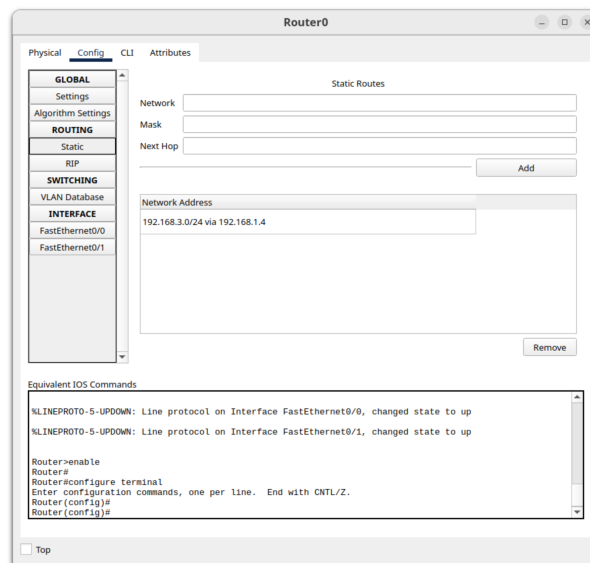


Figure 21: Static Routing

Dynamic Routing Configuration (RIP)

Step 1: Enable RIP

1. Access global configuration mode and start the RIP process:

```
configure terminal
router rip
version 2
```

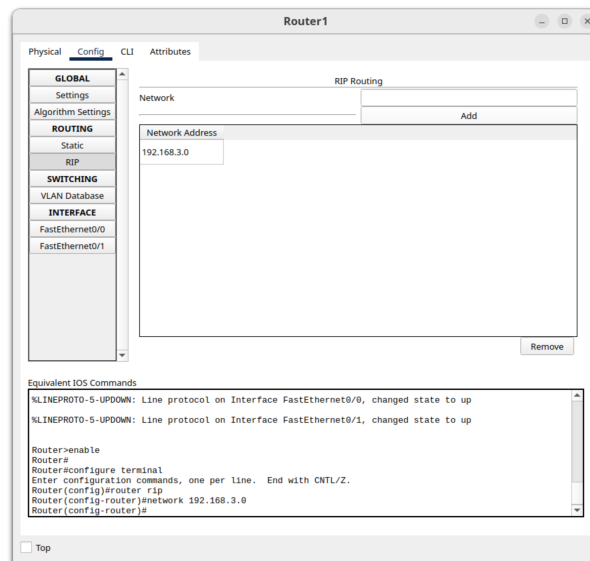


Figure 22: Static Routing

Step 2: Advertise Networks

1. Use the 'network' command to advertise connected networks:

```
network 192.168.1.0
network 10.1.1.0
```

2. Exit RIP configuration mode:

```
exit
```

Step 3: Verify RIP Configuration

1. Use the following commands to verify RIP:

```
show ip protocols  
show ip route
```

Verification and Testing

- Test connectivity using the 'ping' command between devices on different networks.
- For static routes, ensure all networks are reachable.
- For dynamic routes, verify automatic discovery of remote networks.

Conclusion:

Observed Static and Dynamic Routing.

Experiment 17: Configuring Ethernet and Serial Interfaces

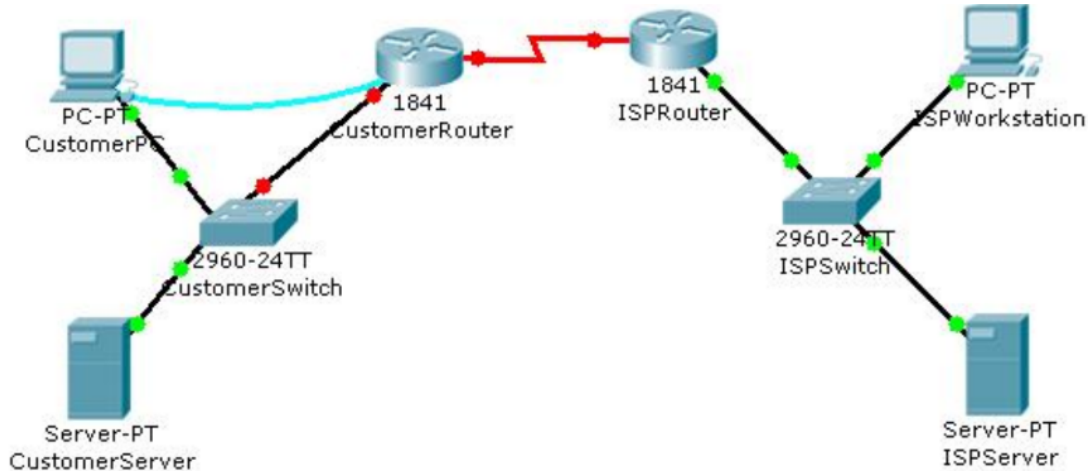


Figure 23: Configuring Ethernet and Serial Interfaces

Objectives:

- Configure a LAN Ethernet interface.
- Configure a WAN serial interface.
- Verify the interface configurations.

Background / Preparation:

In this activity, you will configure the LAN Ethernet interface and the WAN serial interface on the Cisco 1841 router.

Step 1: Configure the LAN Ethernet interface

- a. Use the terminal emulation software on the Customer PC to connect to the Cisco 1841 Router. Enter `cisco` for the console password.
- b. Enter privileged EXEC mode using `cisco123` for the privileged EXEC password. The `Router#` prompt indicates that you are in privileged EXEC mode.
- c. Enter global configuration mode. The `Router(config)#` prompt indicates that you are in global configuration mode.

- d. Identify which LAN interface to configure with an IP address. To configure the Fast Ethernet interface, use the following command:

```
Router(config)#interface FastEthernet 0/0
```

- e. Add a description to the interface:

```
Router(config-if)#description Connected to Switch
```

- f. Specify the IP address and subnet mask for the interface:

```
Router(config-if)#ip address 192.168.1.1 255.255.255.0
```

- g. Ensure that the interface is enabled:

```
Router(config-if)#no shutdown
```

- h. Exit interface configuration mode:

```
Router(config-if)#end
```

Step 2: Verify the LAN interface configuration

Use the `show ip route` command to verify your configuration. This is a partial example of the output:

```
Router#show ip route
```

```
    output omitted;
```

```
Gateway of last resort is not set
```

```
C 192.168.1.0/24 is directly connected, FastEthernet0/0
```

Step 3: Configure the WAN serial interface

Refer to the diagram in the Packet Tracer workspace area and the commands used in Step 1 to configure the WAN serial interface on Customer Router.

Tip: Remember the Cisco IOS CLI Help commands to configure the interface.

- Enter global configuration mode.
- Identify the serial interface to configure.
- Describe the interface. (Connected to ISP)
- Specify the interface IP address and subnet mask:

```
ip address 209.165.200.225 255.255.255.224
```

- Ensure that the interface is enabled.
- End interface configuration mode.

Step 4: Verify the interface configurations

Use the **show run** command to verify your configuration. This is a partial example of the output:

```
Router#show run

...
!
interface FastEthernet0/0
description Connected to Switch
ip address 192.168.1.1 255.255.255.0
duplex auto
speed auto
!
interface Serial0/1/0
description Connected to ISP
ip address 209.165.200.225 255.255.255.224
!
```

Use the **ping** command to verify connectivity to the WAN interface on the ISP router. This is a partial example of the output:

```
Router#ping 209.165.200.226
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 35/37/47 ms
```

Use the **ping** command to verify connectivity to the customer switch. This is a partial example of the output:

```
Router#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/12 ms
```

Step 5: Save the configuration

- a. In privileged EXEC mode, save the running configuration to the startup configuration:

```
Router#copy run start
```

- b. Click the Check Results button at the bottom of this instruction window to check your work.

Experiment 18: Implementing an IP Addressing Scheme

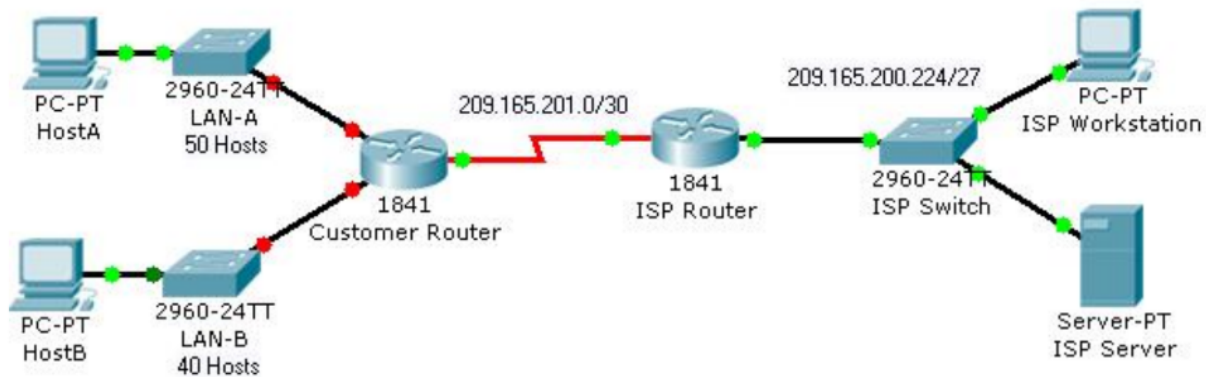


Figure 24: Implementing an IP Addressing Scheme

Objectives:

- Subnet an address space based on the host requirements.
- Assign host addresses to devices.
- Configure devices with IP addressing.
- Verify the addressing configuration.

Background / Preparation:

In this activity, you will subnet the private address space 192.168.1.0/24 to provide enough host addresses for the two LANs attached to the router. You will then assign valid host addresses to the appropriate devices and interfaces. Finally, you will test connectivity to verify your IP address implementation.

Step 1: Subnet an Address Space Based on the Host Requirements

1. You are given the private address space 192.168.1.0/24. Subnet this address space based on the following requirements:
 - LAN-A needs enough addresses for 50 hosts.
 - LAN-B needs enough addresses for 40 hosts.
2. Calculate the subnet mask and allocate IP address ranges:
 - For LAN-A: 192.168.1.0/26 (supports up to 62 hosts).
 - For LAN-B: 192.168.1.64/26 (supports up to 62 hosts).

Step 2: Configure Devices with IP Addressing

Host Configuration

1. Configure **HostA** with the following settings:

- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.1.62

2. Configure **HostB** with the following settings:

- IP Address: 192.168.1.65
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.1.126

Router Configuration

1. Configure the **FastEthernet0/0** interface for LAN-A:

- IP Address: 192.168.1.62
- Subnet Mask: 255.255.255.192
- Port Status: **On**

2. Configure the **FastEthernet0/1** interface for LAN-B:

- IP Address: 192.168.1.126
- Subnet Mask: 255.255.255.192
- Port Status: **On**

Step 3: Verify the Addressing Configuration

1. Test connectivity between **HostA**, **HostB**, **ISP Workstation**, and **ISP Server**.
2. Use the **Add Simple PDU** tool to create pings between the devices.
3. Alternatively, use the command prompt on **HostA** or **HostB** to issue ping commands.
4. Verify successful connectivity:
 - A successful ping indicates proper IP addressing and configuration.
 - Use the **Connectivity Tests** tab to ensure all tests pass.

Conclusion:

Implemented IP Addressing Scheme.

Experiment 19: Configuring Static and Default Routes

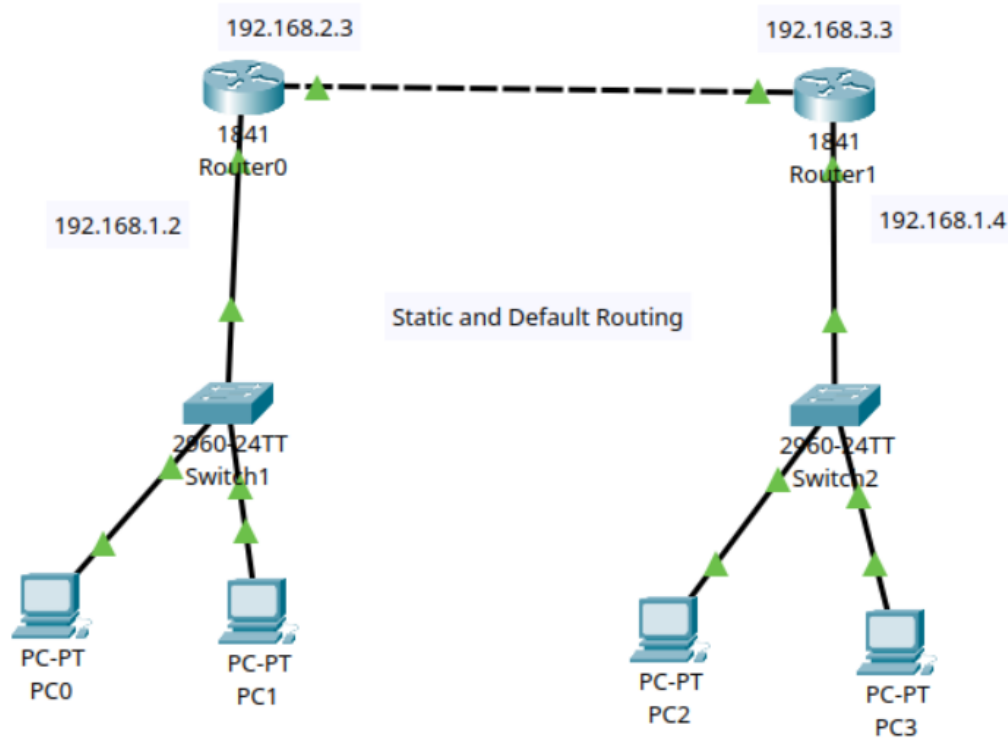


Figure 25: Configuring Static and Default Routes

Objectives:

- Configure static routes on each router to allow communication between all clients.
- Test connectivity to ensure that each device can fully communicate with all other devices.

Background / Preparation:

This topology represents a small WAN. Each device in this network has been configured with IP addresses; however, no routing has been configured. The company management wants to use static routes to connect the multiple networks.

Step 1: Test connectivity between the PCs and the default gateway

To determine if there is connectivity from each PC to its configured gateway, first use a simple ping test.

- a. Click BOpC and go to Desktop > Command Prompt.
- b. From the command prompt, type the `ipconfig` command. Note the IP address for BOpC and the default gateway address. The default gateway address is the IP address for the Fast Ethernet interface on BranchOffice.
- c. Ping 192.168.1.1, the default gateway address for the BranchOffice LAN, from the command prompt on BOpC. This ping should be successful.
- d. Click PNpC and go to Desktop > Command Prompt.
- e. From the command prompt, type the `ipconfig` command. Note the IP address for PNpC and the default gateway address. The default gateway address is the IP address for the Fast Ethernet interface on PartnerNet.
- f. Ping 192.168.3.1, the default gateway address for the PartnerNet LAN, from the command prompt on PNpC. This ping should be successful.
- g. Repeat steps a, b, and c for MOpC and its respective default gateway, the Fast Ethernet interface on MainOffice. Each of these ping tests should be successful.

Step 2: Ping between routers to test connectivity

Use a console cable and terminal emulation software on BOpC to connect to BranchOffice.

- a. Test connectivity with MainOffice by pinging 10.10.10.1, the IP address of the directly connected serial 3/0 interface. This ping should succeed.
- b. Test connectivity with MainOffice by pinging 10.10.10.5, the IP address of the serial 2/0 interface. This ping should fail.
- c. Issue the `show ip route` command from the terminal window of BOpC. Note that only directly connected routes are shown in the BranchOffice routing table. The ping to 10.10.10.5 failed because the BranchOffice router has no routing table entry for 10.10.10.5.
- d. Repeat steps a through c on the other two PCs. The pings to directly connected networks will succeed. However, pings to remote networks will fail.
- e. What steps must be taken to reach all the networks from any PC in the activity?

Step 3: Viewing the routing tables

You can view routing tables in Packet Tracer using the Inspect tool. The Inspect tool is in the Common Tools bar to the right of the topology. The Inspect tool is the icon that appears as a magnifying glass.

- a. In the Common Tools bar, click on the Inspect tool.
- b. Click the MainOffice router and choose **Routing Table**.

- c. Click the BranchOffice router and choose **Routing Table**.
- d. Click the PartnerNet router and choose **Routing Table**.
- e. Move the routing table windows around so that you can see all three at once.
- f. What networks do each of the routers already know about?
- g. Does each router know how to route to all networks in the topology? After comparing the routing tables, close the window for each routing table by clicking the **x** in the upper right corner of each window.

Step 4: Configure default routes on the BranchOffice and PartnerNet routers

To configure static routes for each router, first determine which routes need to be added for each device. For the BranchOffice and the PartnerNet routers, a single default route allows these devices to route traffic for all networks not directly connected. To configure a default route, you must identify the IP address of the next hop router, which in this case is the MainOffice router.

- a. From the Common toolbar, click the Select tool.
- b. Move the cursor over the red serial link between the BranchOffice router and the MainOffice router. Notice that the interface of the next hop is S3/0.
- c. Move the cursor over the MainOffice router and note that the IP address for Serial 3/0 is 10.10.10.1.
- d. Move the cursor over the red serial link between the PartnerNet router and the MainOffice router. Notice that the interface of the next hop is S2/0.
- e. Move the cursor over the MainOffice router and note that the IP address for Serial 2/0 is 10.10.10.5.
- f. Configure the static routes on both the BranchOffice and PartnerNet routers using the CLI. Click the BranchOffice router, and click the CLI tab.
- g. At the **BranchOffice>** prompt, type **enable** to enter privileged EXEC mode.
- h. At the **BranchOffice#** prompt, type **configure terminal**.
- i. The syntax for a default route is **ip route 0.0.0.0 0.0.0.0 next_hop_ip_address**.
Type:


```
ip route 0.0.0.0 0.0.0.0 10.10.10.1
```
- j. Type **end** to get back to the **BranchOffice#** prompt.
- k. Type **copy run start** to save the configuration change.
- l. Repeat steps f through k on the PartnerNet router, using 10.10.10.5 as the next hop IP address.

Step 5: Configure static routes at Main Office

The configuration of static routes at the Main Office is a bit more complex because the MainOffice router is responsible for routing traffic to and from the Branch Office and PartnerNet LAN segments. The MainOffice router knows only about routes to the 10.10.10.0/30, 10.10.10.4/30, and 192.168.2.0/24 networks because they are directly connected. Static routes to the 192.168.1.0/24 and 192.168.3.0/24 networks need to be added so that the MainOffice router can route traffic between the networks behind the BranchOffice and PartnerNet routers.

- a. Click the MainOffice router, and then click the CLI tab.
- b. At the MainOffice> prompt, type `enable` to enter privileged EXEC mode.
- c. At the MainOffice# prompt, type `configure terminal`.
- d. The syntax for a static route is `ip route network subnet_mask next_hop_ip_address`:

```
ip route 192.168.1.0 255.255.255.0 10.10.10.2
ip route 192.168.3.0 255.255.255.0 10.10.10.6
```
- e. Type `end` to return to the MainOffice# prompt.
- f. Type `copy run start` to save the configuration change.
- g. Repeat steps a through e from Step 3. View the routing tables and notice the difference in the routing tables. The routing table for each router should have an “S” for each static route.

Step 6: Test connectivity

Now that each router in the topology has static routes configured, all hosts should have connectivity to all other hosts. Use ping to verify connectivity.

- a. Click BOpC and click the Desktop tab.
- b. Choose the Command prompt option.
- c. Type `ping 192.168.3.2`. The ping should be successful, verifying that the static routes are configured properly.
- d. Type `ping 192.168.2.2`. Notice that the result is successful even though you did not specifically add the 192.168.2.0 network as a static route into any of the routers. Because a default route was used on the BranchOffice and PartnerNet routers, a route for the 192.168.2.0 network was not needed. The default route sends all traffic destined off network to the MainOffice router. The 192.168.2.0 network is directly connected to the MainOffice router; therefore, no additional routes needed to be added to the routing table.
- e. Click the Check Results button at the bottom of this instruction window to check your work.

Experiment 20: Configuring RIP

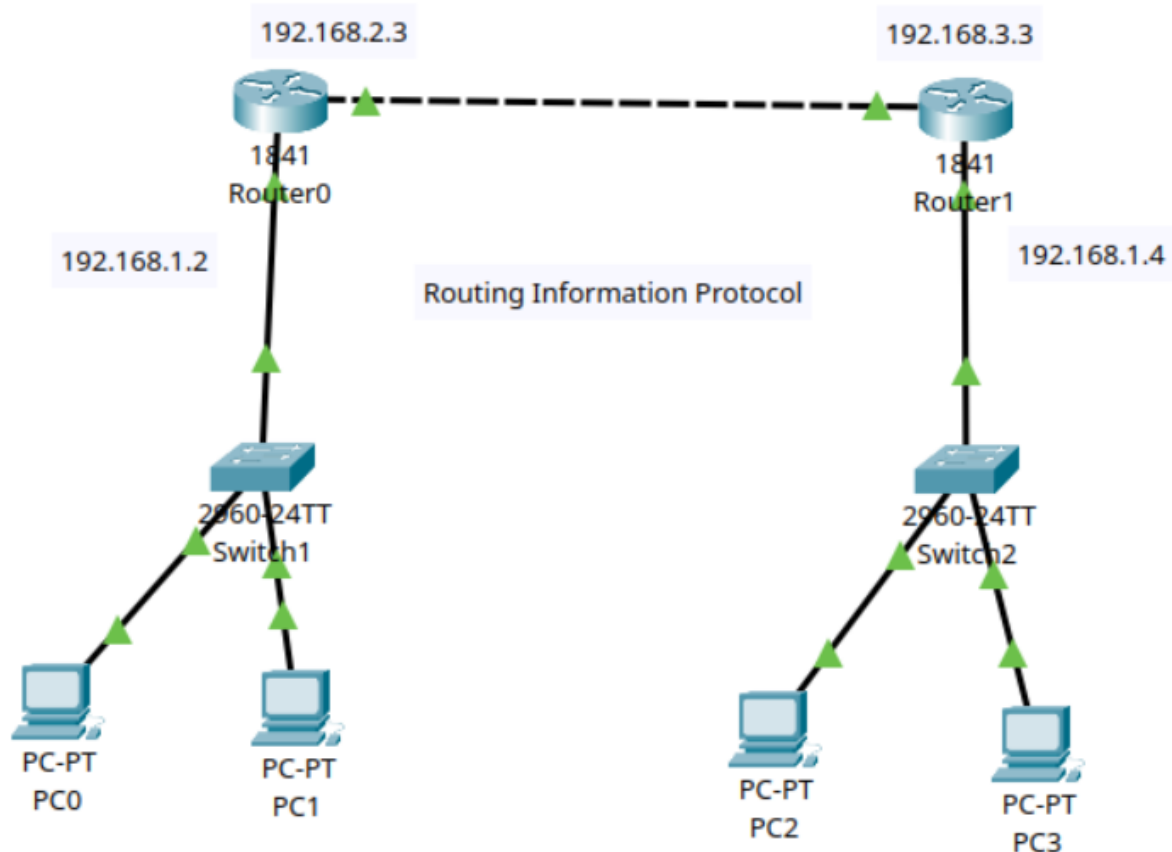


Figure 26: Routing Information Protocol [RIP]

Objectives:

- Configure routers using basic interface configuration commands.
- Enable RIP.
- Verify the RIP configuration.

Background / Preparation:

A simple routed network has been set up to assist in reviewing RIP routing behavior. In this activity, you will configure RIP across the network and set up end devices to communicate on the network.

Step 1: Configure the SVC01 router and enable RIP

- a. From the CLI, configure interface Fast Ethernet 0/0 using the IP address 10.0.0.254 /8.
- b. Configure interface serial 0/0/0 using the first usable IP address in network 192.168.1.0 /24 to connect to the RTR01 router. Set the clock rate at 64000.
- c. Configure interface serial 0/0/1 using the first usable IP address in network 192.168.2.0 /24 with a clock rate of 64000.
- d. Using the **no shutdown** command, enable the configured interfaces.
- e. Configure RIP to advertise the networks for the configured interfaces.
- f. Configure the end devices:
 - Server0 uses the first usable IP address in network 10.0.0.0 /8. Specify the appropriate default gateway and subnet mask.
 - Printer0 uses the second usable IP address in network 10.0.0.0 /8. Specify the appropriate default gateway and subnet mask.

Step 2: Configure the RTR01 router and enable RIP

- a. Configure interface Fast Ethernet 0/0 using the first usable IP address in network 192.168.0.0 /24 to connect to the RTR02 router.
- b. Configure interface serial 0/0/0 using the second usable IP address in network 192.168.1.0 /24 to connect to the SVC01 router.
- c. Configure interface Fast Ethernet 0/1 using the IP address 172.16.254.254 /16.
- d. Using the **no shutdown** command, enable the configured interfaces.
- e. Configure RIP to advertise the networks for the configured interfaces.
- f. Configure the end devices:
 - PC0 uses the first usable IP address in network 172.16.0.0 /16.
 - PC1 uses the second usable IP address in network 172.16.0.0 /16.
 - Specify the appropriate default gateway and subnet mask on each PC.

Step 3: Configure the RTR02 router and enable RIP

- a. Configure interface Fast Ethernet 0/0 using the second usable IP address in network 192.168.0.0 /24 to connect to the RTR01 router.
- b. Configure interface serial 0/0/0 using the second usable IP address in network 192.168.2.0 /24 to connect to the SVC01 router.

- c. Configure interface Fast Ethernet 0/1 using the IP address 172.17.254.254 /16.
- d. Using the `no shutdown` command, enable the configured interfaces.
- e. Configure RIP to advertise the networks for the configured interfaces.
- f. Configure the end devices:
 - PC2 uses the first usable IP address in network 172.17.0.0 /16.
 - PC3 uses the second usable IP address in network 172.17.0.0 /16.
 - Specify the appropriate default gateway and subnet mask on each PC.

Step 4: Verify the RIP configuration on each router

- a. At the command prompt for each router, issue the commands `show ip protocols` and `show ip route` to verify RIP routing is fully converged. The `show ip protocols` command displays the networks the router is advertising and the addresses of other RIP routing neighbors. The `show ip route` command output displays all routes known to the local router, including the RIP routes, which are indicated by an “R”.
- b. Every device should now be able to successfully ping any other device in this activity.
- c. Click the Check Results button at the bottom of this instruction window to check your work.

Conclusion:

Configured Routing Information Protocol